# Rogelio Leonardo Mendez Macias

**Embedded Computer Vision & Edge AI Engineer**

Perception · ADAS · NPU Deployment

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## Summary


Computer vision engineer who turns perception models into measurable on-device systems. Built and validated an end-to-end edge inference pipeline on Hailo-8 NPU + Raspberry Pi 5 (PyTorch → ONNX → INT8 → HEF), published in Springer LNCS. Applied experience in object detection, license plate recognition/OCR, segmentation and temporal risk modeling on real-world dashcam data. First-place winner of an IBM-judged national AI hackathon.


## Projects

**Embedded Vehicle Perception & License Plate Recognition on NPU — Thesis (UAM Azcapotzalco)** 2025–2026

- Deployed on-device inference on Raspberry Pi 5 + Hailo-8 NPU (PyTorch training → ONNX export → INT8 quantization → HEF compilation with Hailo Dataflow Compiler): 30–58 FPS on-device, 230 ms end-to-end latency, 88% model compression.
- Architected perception across three layers: vehicle detection on the edge NPU, license plate OCR on a central layer (>93% precision) and an LLM+TTS cognitive layer, integrated as structured events over an MQTT backbone; zero operational failures during validation.
- Owned the full lifecycle: dataset curation, training, export, quantization, deployment and validation. Published in Springer LNCS (MCPR 2026). Demo: youtu.be/n2k4fdbCWAk | Code: github.com/Rogelio756/camaxtli-adas/tree/feat/copiloto/nodes/copiloto

**AI-Powered Road Risk Intelligence — Samsung Innovation Campus** 2025–2026

- Co-developed, in a 3-person team, a 5-stage perception pipeline on real-world driving data: YOLOv8n detection + SegFormer-B0 segmentation + GCNEncoder + RiskLSTM + BNN/MC Dropout uncertainty estimation.
- Designed and validated the RiskLSTM strategy individually: Pearson $r = 0.916$ on 61,345 BDD100K images + 925 real-world BDDA dashcam clips.
- Coordinated three pipeline strategies. Code: github.com/Rogelio756/Equipo8-Grupo8-SIC-2025-/tree/estrategia-1

**FloodSense — Real-Time Hydrological Intelligence** Jun 2026

- Won 1st place at Hackathon Concienc.IA 2026 (Young AI Leaders CDMX Hub, hosted at Tec de Monterrey CCM; IBM-judged) with a 3-agent ML system built in 48 h: XGBoost classifier (ROC-AUC 0.86) + risk regressor ($R^2 = 0.999$) + IBM watsonx/Granite alert layer.
- Trained on 591,706 CONAGUA records (1877–2024), covering 532 zones across 16 CDMX boroughs with 6 h anticipation. Selected for R&D incubation by ADIP (Mexico City's Digital Agency for Public Innovation).
- Live: floodsensetec.vercel.app | Code: github.com/Rogelio756/hackaton_agentes_concienc_ia

**IoT Plant-Care System for Horticulture — Samsung Innovation Campus** 2024

- Built an IoT horticulture system with an IPN team; submitted to the Santander Reto Universitario.

## Experience

**Grupo Modelo (AB InBev)** — Data Engineering & Process Digitalization Intern May–Nov 2023

- Built Python + SQL automation pipelines that reduced reporting workload by >5 hrs/week.
- Initiated a PPE-detection computer vision prototype for industrial safety, beyond assigned scope.

## Publication

Méndez-Macías, R.L., Villegas-Cortez, J., Ferreyra Ramírez, A., Zúñiga-López, A., Cordero-Sánchez, S. "Embedded System for Vehicle Environment Perception and License Plate Recognition (LPR) Using Computer Vision and Deep Learning." In: *Pattern Recognition. MCPR 2026*, Lecture Notes in Computer Science, vol. 16623, pp. 247–258. Springer, Cham (2026). DOI: 10.1007/978-3-032-28393-1_22

## Awards & Talks

- **1st Place — Hackathon Concienc.IA 2026** (FloodSense; IBM-judged) Jun 2026
- **Presenter — MCPR 2026**, 18th Mexican Conference on Pattern Recognition (INAOE), Ciudad Juárez Jun 2026
- **Finalist — Falling Walls Lab**, Mexico City final (ADAS project) 2026
- **MUTVI Recognition**, UAM — oral talk "El copiloto que nunca duerme" (youtu.be/nesc0EuF2kk) 2026
- **Presenter — NEO International Congress** (distributed AI architectures, real-time perception) 2025
- **Executive Pitch — Santander Reto Universitario**, presented to Samsung senior leadership 2024

## Technical Skills

**Edge AI & Model Optimization:** PyTorch, PyTorch Geometric, YOLOv8, SegFormer, OpenCV, ONNX, INT8 quantization, Hailo-8 NPU / Dataflow Compiler (HEF), CUDA
**Embedded Systems & Integration:** Raspberry Pi 5 (Linux), MQTT, REST APIs, Docker (basic), Git/GitHub
**Data & ML:** Python (advanced), NumPy, Pandas, XGBoost, SQL, IBM watsonx/Granite, MATLAB, LaTeX; control theory (PID)

## Education

**Universidad Autónoma Metropolitana — Azcapotzalco**, Mexico City Expected: Apr 2027
B.Sc. in Electronics Engineering · Specialization: Edge AI, Computer Vision, Embedded Systems

**Samsung Innovation Campus** — AI & IoT Engineering Program (2024) · AI & Digital Skills (2025–2026)

**Languages:** Spanish (native) · English (A2; presented MCPR 2026 talk in English, reads technical documentation)